

CLASS-8

LESSON-1 RATIONAL NUMBERS

(This PDF Based on NCERT Book)

RATIONAL NUMBER(परिमेय संख्या)-A rational number is simply a number that can be written as a **fraction** $\frac{p}{q}$, where:

- p and q are **integers** (whole numbers, including negative ones and zero).
- The denominator, q, is **not zero** ($q \neq 0$).

PROPERTIES OF RATIONAL NUMBERS(परिमेय संख्याओं के गुण)-

1. Closure Property(समापन गुण)

When you do an operation (add, subtract, or multiply) on any two rational numbers, the answer will **always be another rational number**. The number set is "closed," meaning you can't escape it by using those operations.

- **Addition:** $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$ (Rational)
- **Subtraction:** $\frac{3}{4} - \frac{1}{4} = \frac{2}{4}$ (Rational)
- **Multiplication:** $\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$ (Rational)
- **Division is NOT closed** because you cannot divide by zero ($\frac{5}{0}$ is undefined)

2. Commutative Property(क्रमचयी गुण)

For addition and multiplication, you can **change the order** of the numbers without changing the final result.

- **Addition:** $a + b = b + a$ (Example: $2 + 5 = 5 + 2$)
- **Multiplication:** $a \times b = b \times a$ (Example: $2 \times 5 = 5 \times 2$)
- **Note:** This **does not work** for subtraction ($5 - 2 \neq 2 - 5$) or division ($4 \div 2 \neq 2 \div 4$).

3. Associative Property(साहचर्य गुण)

For addition and multiplication, when you have three or more numbers, you can **change the grouping** (which numbers you calculate first) without changing the final result.

- **Addition:** $(a + b) + c = a + (b + c)$ (Example: $(1+2)+3 = 1+(2+3) \rightarrow 3+3 = 1+5 \rightarrow 6=6$)
- **Multiplication:** $(a \times b) \times c = a \times (b \times c)$

- **Note:** This **does not work** for subtraction or division.

4. The role of 0 (0 की भूमिका)

- **Zero (0)** is the number you can **add** to any rational number without changing its value.
- $a + 0 = a$ (Example: $5 + 0 = 5$)

5. The role of 1 (1 की भूमिका)

- **One (1)** is the number you can **multiply** any rational number by without changing its value.
- $a \times 1 = a$ (Example: $5 \times 1 = 5$)